AI-Driven Commercial Vehicle Safety and Violation Detection System Using Vision and GIS Context

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1. Project Title: AI-Driven Commercial Vehicle Safety and Violation Detection System Using Vision and GIS Context

2. Abstract

Commercial vehicle collisions frequently result from driver fatigue, operator distraction, inappropriate approach speed along curved road segments, and passenger overloading. These risks are compounded across high-risk transportation corridors and mountainous terrains, where complex road geometry, steep gradients, and restricted visibility demand continuous operational vigilance. While conventional telematics solutions capture basic positional coordinates and velocity metrics for retrospective post-incident reporting, they lack the contextual, real-time decision logic required to proactively identify hazard conditions prior to impact. Furthermore, existing commercial fleet systems rarely correlate vehicular velocity against the physical geometry of upcoming road infrastructure.

To address these limitations, this project presents an embedded edge-computing safety architecture that integrates multi-camera video feeds with Geographic Information System (GIS) spatial data to monitor transit conditions and enforce regulatory compliance in real time. The data acquisition framework synchronizes inputs from three USB/CSI optical sensors alongside a standard GNSS/GPS positioning receiver. A driver-facing camera utilizes MediaPipe for facial landmark extraction to detect eyelid closure, yawning, and driver distraction, and Ultralytics YOLO to detect mobile phone usage; an interior cabin camera applies Ultralytics YOLO with ByteTrack to track occupant count and identify passenger overcrowding; and a forward-facing camera detects vehicles and obstacles ahead and records road-scene evidence for incident review. All video from the driver-facing and interior cameras is processed entirely on-device, analyzed in memory and discarded immediately after inference, with only event-level metadata and short, violation-triggered evidence clips retained raw video is never transmitted off the vehicle. Concurrently, the vehicle coordinates are matched against OpenStreetMap layers and SRTM elevation models within PostgreSQL/PostGIS to evaluate curvature radii, gradients, and turn proximity.

A local inference engine implemented in Python evaluates these spatial and visual parameters using speed-safety logic independently developed by the team, drawing on the behavioral principles described in the referenced institutional patent, to compute dynamic safe-speed thresholds and flag non-compliant velocity profiles. In tunnels and mountainous terrain, where GNSS signal is frequently lost, the system falls back to inertial dead-reckoning estimating position and heading from IMU data and camera-derived speed against locally cached map tiles so curvature and gradient checks continue for a bounded grace period before the system flags reduced positional confidence to the driver. A violation arbitrator filters sensor noise through multi-frame confirmation and confidence-weighted debouncing, preventing single noisy or momentary detections from triggering false alerts, issues in-cabin alerts, and buffers telemetry logs locally for opportunistic synchronization via MQTT and REST/HTTP APIs over cellular

networks to a React.js, Leaflet.js, and Chart.js fleet dashboard. Deployed on an edge AI module (Raspberry Pi 5 / NVIDIA Jetson Orin Nano), the system operates autonomously offline, delivering a cost-effective monitoring platform without requiring invasive vehicle modifications. System performance will be evaluated on recorded video datasets and simulated drive routes to measure detection precision and recall, false-alarm rate, speed-violation detection accuracy, and real-time inference latency on edge hardware.

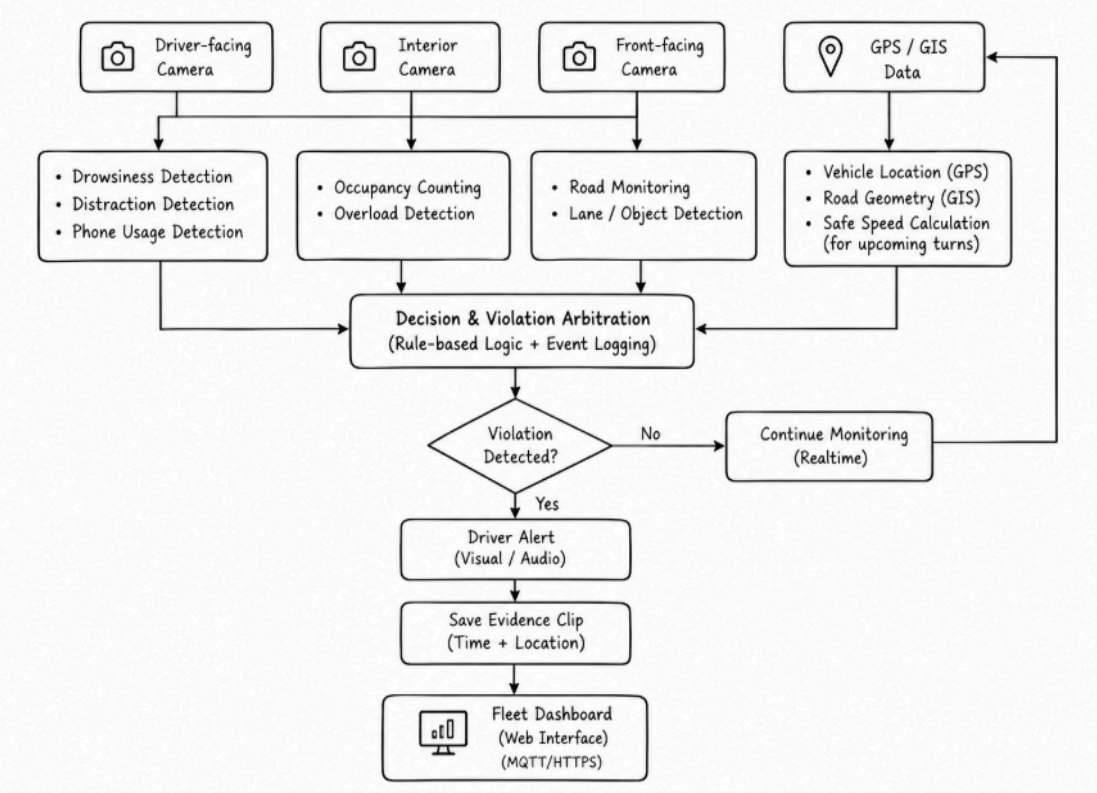
3. Approach Details

- **Multi-Camera and Telemetry Ingestion:** Captures concurrent video streams from a driver-facing camera, an interior cabin camera, and a forward-facing road camera, alongside continuous latitude, longitude, and velocity coordinates via a GPS/GNSS receiver. Frame capture across the three cameras is timestamp-aligned via a shared system clock for synchronized multi-sensor fusion.
- **Edge-Based Computer Vision Analytics with On-Device Privacy:** Executes lightweight deep learning models locally to track facial landmarks for driver drowsiness and distraction, detect mobile phone usage, detect vehicles and obstacles ahead, and estimate occupant count. Video frames from the driver-facing and interior cameras are processed in memory and discarded immediately after inference; only derived event metadata and short, violation-triggered clips are stored, encrypted at rest.
- **GIS Spatial & Curvature Correlation:** Queries a locally cached PostGIS instance (from OpenStreetMap and SRTM data) to retrieve curvature radii, junction proximity, and gradient, correlating them with the vehicle's incoming speed vector without depending on a live remote query.
- **GPS-Denied Fallback for Tunnels and Mountainous Terrain:** When GNSS fix is lost or its accuracy estimate exceeds threshold, the system estimates position and heading via IMU dead-reckoning combined with camera-derived speed, continues map-matching against cached tiles for a bounded grace window (e.g., 15–30 seconds), and enters a conservative "reduced-confidence" mode until GNSS fix is reacquired.
- **Patent-Informed Speed Determination Engine:** Implements dynamic, context-appropriate speed thresholds using logic independently authored by the team based on the referenced institutional patent's behavioral principles.
- **Noise-Resilient Violation Arbitration:** Requires multi-frame confirmation and confidence-weighted debouncing before escalating a detection to a logged violation, cross-checking against an independent signal where available; degraded or unavailable sensor feeds suppress their dependent alerts rather than triggering false ones.
- **Telemetry Dispatch:** Packages arbitrated incidents, sensor states, and short evidence clips into JSON/MQTT packets for opportunistic cellular transmission.
- **Supervisory Fleet Dashboard:** Real-time geospatial tracking, automated violation reports, and role-based access on a React.js/Leaflet.js/Chart.js console.

4. Relationship to the Institute Patent

This project draws direction from the institute's patent *"System for Monitoring Active Safety of a Vehicle and Method Thereof"* (Complete Specification dated 25 April 2024), which establishes the concept of AI-based active safety using interior/exterior imaging, vehicle sensors, curve anticipation and optimal-speed advice. The specification describes the system's objectives, data flow and claims but does not disclose specific algorithms, models or thresholds. All detection models, the speed-advisory logic, the violation-arbitration rules and the dashboard in this project are designed and implemented independently by the project team for this mini project, using the patent as a conceptual reference rather than as source code or a fixed formula.

5. Architecture Diagram



6. Use Cases

- **Commercial Freight and Logistics Fleets:** Monitoring long-haul cargo drivers for fatigue, mobile phone distraction, and adherence to safe approach speeds around sharp highway turns.
- **Public Transit and State Bus Services:** Enforcing legal passenger load limits to prevent dangerous overcrowding and tracking driving behavior across scheduled urban/intercity routes.

- **Government Transport Authorities & Enforcement Boards:** Providing Transport Commissioners with automated, evidence-backed violation logs to support regulatory auditing and targeted compliance interventions.
- **Hazardous Cargo Transport Operations:** Continuous spatial speed governance and vigilance verification through high-risk geographical sectors, mountain passes, and dense urban crossings.

7. **Technology Stack**

Category	Technologies / Tools Used
Hardware	Edge AI Module (Raspberry Pi 5 / NVIDIA Jetson), USB/CSI Camera Units, GNSS/GPS Receiver
Programming Languages	Python, JavaScript
Frameworks & Libraries	OpenCV, MediaPipe, Ultralytics YOLO, ByteTrack, FastAPI
GIS & Database	PostgreSQL with PostGIS, OpenStreetMap data, SRTM elevation data
Protocols	MQTT (Mosquitto broker), REST/HTTP
Security	AES-256 at-rest encryption for evidence clips, role-based dashboard access
Visualization & UI	React.js, Leaflet.js, Chart.js

8. VETS Justification

V - Viability: The project uses open-source tools (OpenCV, MediaPipe, YOLO, ByteTrack, PostGIS) and free OpenStreetMap and SRTM data. Public datasets (DMD, YawDD, CrowdHuman, India Driving Dataset) support the vision modules. It can be prototyped on a laptop with recorded video and GPS traces, following a phased rollout from a single-camera MVP to full multi-sensor integration to manage implementation risk, then moved to a Raspberry Pi 5 or NVIDIA Jetson Orin Nano. The team's skills in AI/ML, IoT and full-stack development cover all modules.

E - Engineering Depth: The work covers real-time facial-landmark and object detection with on-device privacy handling, person tracking, GPS/IMU-fused map matching with curve and gradient computation for GNSS-denied operation, an original speed-safety implementation informed by the referenced institutional patent, noise-resilient multi-sensor violation arbitration, and integration of synchronised sensor streams on edge hardware with offline storage and MQTT sync.

T - Trend Alignment: The project applies AI and computer vision for perception, edge computing for on-vehicle inference in low-connectivity regions, IoT for sensor integration, and GIS-based analytics for context-aware decisions.

S - Social / Industrial Impact: It warns drivers before violations become accidents, improving road safety in hilly regions such as Jammu and Kashmir. Explicit on-device privacy handling, GNSS-outage resilience, and noise-filtered alerting make the system credible for real fleet deployment rather than only controlled conditions. It gives fleets and transport authorities automated, evidence-backed monitoring on low-cost hardware without vehicle modifications, suiting freight, bus and hazardous-cargo operations.

9. Risk and Mitigation

- **Multi-camera and GIS integration risk:** the team will build and test the driver-camera module first, then the cabin camera, then the forward camera, integrating one stream at a time rather than all at once.
- **Real-time performance on edge hardware:** initial development and benchmarking will run on a laptop with recorded video; only after each module meets its target latency will it move to the Raspberry Pi 5 or Jetson Orin Nano.
- **GIS as extension, not dependency:** the core patent-inspired speed-advisory logic (curve anticipation from a trained route model) works without GPS or GIS data, so the system remains functional if the GIS layer is delayed or dropped.
- **Fallback scope:** if full sensor fusion is not achievable within the timeline, the deliverable will prioritise driver monitoring and curve-based speed advice as the demonstrated core, with passenger counting and GIS context as stretch goals.

10. References

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